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AgentDropout: Dynamic Redundancy Elimination for Multi-Agent Collaboration Efficiency

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***Abstract*—Multi-agent systems powered by large language models (LLMs) have demonstrated potential for collaborative problem-solving, yet increasing the number of agents often introduces redundant reasoning and communication overhead, sometimes degrading performance. We propose AgentDropout, a dynamic strategy inspired by dropout regularization in neural networks, which selectively deactivates low-contribution agents during multi-agent collaboration. At each round of discussion, AgentDropout computes a *semantic novelty score* for every agent by measuring the divergence of its output relative to the current group consensus. Agents whose novelty score falls below an adaptive threshold are temporarily deactivated, reducing token consumption without sacrificing viewpoint diversity. We evaluate AgentDropout on mathematical reasoning (GSM8K), commonsense reasoning (StrategyQA), and collaborative code generation (HumanEval) tasks. Across three independent runs, AgentDropout achieves accuracy comparable to or modestly above fixed 5-agent debate while reducing total token consumption by 38.0–43.5% and debate rounds by 20.5% on average. Pareto analysis reveals a promising efficiency–quality trade-off, suggesting that dynamic agent deactivation may be useful for deploying multi-agent LLM systems under computational budget constraints.**

***Keywords*—Multi-agent systems, large language models, efficiency optimization, semantic novelty, dropout**

I. Introduction

Large language models (LLMs) have been increasingly organized into multi-agent systems to tackle complex reasoning and generation tasks [1]. Frameworks such as multi-agent debate [2], structured debate with a judge [3], role-based collaboration [4][5], and conversation-driven pipelines [6] have shown performance gains over single-model inference on benchmarks spanning mathematical reasoning, code generation, and commonsense question answering.

However, the benefits of multi-agent collaboration are not unconditional. Smit et al. [7] demonstrate that multi-agent debate systems do not reliably outperform simpler prompting strategies such as self-consistency [8] and often exhibit high sensitivity to hyperparameter settings. A fundamental issue is *redundancy*: as agents converge toward consensus, their outputs become increasingly similar, and continued participation of all agents introduces duplicated reasoning and amplified token costs without proportional accuracy gains. In our own experiments, the 7-agent debate configuration tends to underperform both 3-agent and 5-agent setups. Simply reducing the agent count, however, risks eliminating diverse viewpoints valuable during early deliberation.

We address this tension by drawing an analogy to *dropout regularization* in neural networks [9]. In standard dropout, randomly deactivating neurons during training prevents co-adaptation and encourages robust representations. Analogously, we propose **AgentDropout**, a mechanism that dynamically deactivates agents whose contributions have become redundant relative to the evolving group consensus. Unlike neural-network dropout, which operates stochastically, AgentDropout is *novelty-driven*: at each debate round, it quantifies the *semantic novelty* of every agent's output and deactivates those falling below a threshold.

Our contributions are: (1) We formalize per-agent semantic novelty in the context of multi-agent LLM collaboration and propose a lightweight, threshold-based deactivation mechanism. (2) We conduct experiments on GSM8K [10], StrategyQA [11], and HumanEval [12], demonstrating that AgentDropout achieves competitive accuracy with significantly lower computational cost. (3) We analyze the Pareto frontier between active agents and task accuracy, identifying favorable operating points for budget-constrained deployments.

II. Related Work

Multi-Agent LLM Collaboration. The idea of leveraging multiple LLM instances to improve output quality has gained traction along several axes, including multi-agent debate, structured debate with divergent thinking, fixed role-assignment mimicking organizational workflows, general-purpose multi-agent conversation, and round-table consensus across diverse LLMs [13]. While these approaches demonstrate the promise of multi-agent collaboration, they typically operate with a fixed team composition throughout the task, leaving the question of dynamic agent management largely unexplored.

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Recent work further broadens multi-agent collaboration beyond answer aggregation. Role division and collaborative reasoning have been used to coordinate complementary interpretations in complex contextual tasks [14], while schema-aware language models and cross-statement consistency checking illustrate how structured constraints can stabilize agent outputs [15]. Similar considerations arise in AI-assisted analysis of expressive artifacts [16], where the allocation of specialized perspectives affects the reliability of a collective assessment. Conservative risk-sensitive reinforcement learning [17] and lightweight edge-cloud cooperative inference [18] likewise emphasize that reliable decisions must be paired with resource-aware coordination.

A second line of work focuses on dependable execution under imperfect intermediate signals. Protocol anomaly detection [19], constraint-consistent skill composition [20], and stable fault diagnosis under data imbalance [21] each motivate explicit mechanisms for identifying unreliable states before they propagate through a workflow. Calibrated LLM-as-a-judge scoring [22] and graph-based risk scoring with structural regularization [23] further show the value of combining uncertainty estimates with relational evidence. These perspectives support dynamic management policies that retain diverse reasoning when it remains informative and curtail computation when the marginal signal has weakened.

Efficiency also depends on how agent systems are deployed and maintained. Token-level GPU pooling [24], drift-aware agentic ETL [25], and semantic repair for schema drift [26] characterize resource allocation and adaptation as first-class system concerns. Data-quality-aware multimodal prediction [27] and representation learning for XBRL-guided forecasting [28] reinforce the importance of monitoring the quality of the signals entering a decision process. Together, these studies motivate treating participation as an adaptive computational resource rather than a fixed property of a collaborative team.

Related reliability mechanisms span persistent state, backend routing, and structured reasoning. Evidence-gated memory writing [29], KV-cache-aware routing [30], and XBRL-aware agents with anti-hallucination verification [31] demonstrate complementary controls over what information is retained, retrieved, and trusted. SLO-aware elastic scheduling [32], data-centric sequential modeling [33], multi-feature sentiment modeling [34], correlation-bias mitigation [35], asymmetric-loss capacity planning [36], dependency modeling for cloud-service anomalies [37], and explainable LLM representations [38] collectively underscore the value of calibrated, interpretable, and cost-sensitive control in complex AI systems.

Efficiency in Multi-Agent Systems. Li et al. [39] show that sparse communication topologies—where agents communicate only with a subset of peers—can reduce token costs by over 40% while preserving accuracy. Extensive empirical comparisons further reveal that many debate configurations underperform simpler baselines, motivating more principled agent management. Our work differs from topology-based approaches by dynamically adjusting the *number* of active agents rather than the *structure* of communication.

Dropout as Regularization. Dropout was introduced to prevent overfitting in deep neural networks by randomly zeroing activations during training. Self-consistency decoding and chain-of-thought prompting [40] share the broader principle of sampling diverse reasoning paths and aggregating results. Our work extends the dropout metaphor from the neural-network level to the *agent level*, treating each agent as a computational unit that can be selectively deactivated based on its marginal contribution.

III. Methodology

A. Problem Formulation

Consider a multi-agent system consisting of N agents $\{A_1, A_2, \ldots, A_N\}$ engaged in iterative deliberation over at most T rounds. At round t, every active agent A_i generates an output $o_i^{(t)}$ conditioned on the consensus from the previous round, $\mathcal{C}^{(t-1)}$. After all active agents have produced their outputs, the system constructs an updated consensus $\mathcal{C}^{(t)}$ (see Section III-E). The novelty score $\mathcal{S}_i^{(t)}$ is computed by comparing each agent's output $o_i^{(t)}$ against $\mathcal{C}^{(t)}$. Our goal is to deactivate agents whose outputs are redundant with respect to $\mathcal{C}^{(t)}$, reducing the number of active agents in subsequent rounds without impairing task performance.

B. Semantic Novelty Scoring

For each agent A_i at round t, we compute a semantic novelty score $\mathcal{S}_i^{(t)}$ that quantifies how different its output is from the current consensus. Let $\mathbf{e}_i^{(t)} = \mathrm{Enc}\big(o_i^{(t)}\big)$ be the sentence embedding of agent A_i's output and $\mathbf{e}_{\mathcal{C}}^{(t)} = \mathrm{Enc}\big(\mathcal{C}^{(t)}\big)$ be the embedding of the consensus. We define:

$$\mathcal{S}_i^{(t)} = 1 - \frac{\mathbf{e}_i^{(t)} \cdot \mathbf{e}_{\mathcal{C}}^{(t)}}{\| \mathbf{e}_i^{(t)} \| \, \| \mathbf{e}_{\mathcal{C}}^{(t)} \|} \tag{1}$$

which is one minus the cosine similarity between the agent's output and the consensus. A high $\mathcal{S}_i^{(t)}$ indicates substantially different content from the group summary, while a low value suggests redundancy. We employ the lightweight `all-MiniLM-L6-v2` sentence encoder, adding negligible overhead relative to the LLM generation cost. The scoring stage also draws on evidence-verified reasoning as articulated in VerifyRCA [41]. Rather than interpreting semantic divergence as sufficient evidence of a useful contribution, the score is used as a compact signal for whether an agent should remain available to the collaborative process. This framing keeps deactivation tied to observable differences between an output and the current group state. It also leaves the final task judgment to the consensus mechanism instead of conflating novelty with correctness.

C. Adaptive Deactivation Mechanism

Given the novelty scores, the deactivation decision for each agent is:

$$\mathrm{Active}_i^{(t+1)} = \begin{cases} 1 & \text{if } \mathcal{S}_i^{(t)} \geq \tau^{(t)} \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

where $\tau^{(t)}$ is a round-dependent threshold. To avoid premature elimination of diverse viewpoints, we decay the threshold over rounds:

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$$\tau^{(t)} = \tau_{\text{base}} \cdot \left(1 - \alpha \cdot \frac{t}{T}\right) \tag{3}$$

Here, τ_{base} is a base threshold and $\alpha \in [0,1]$ is a decay rate. We enforce a minimum active agent count $N_{\min} \geq 2$ to prevent degeneration to a single-agent regime. The adaptive rule further incorporates the early-truncation perspective developed for multi-step agents in ErrorChain [42]. A declining novelty trajectory is treated as an indication that additional rounds are increasingly likely to repeat an already established line of reasoning. Threshold decay delays this intervention until the group has had an opportunity to explore competing viewpoints. The minimum-agent constraint preserves a controlled recovery path when an apparently settled trajectory later proves incomplete.

D. Reactivation Under Consensus Shift

To handle situations where the group consensus shifts substantially, we define a consensus shift measure:

$$\Delta^{(t)} = 1 - \frac{\mathbf{e}_{\mathcal{C}}^{(t)} \cdot \mathbf{e}_{\mathcal{C}}^{(t-1)}}{\| \mathbf{e}_{\mathcal{C}}^{(t)} \| \, \| \mathbf{e}_{\mathcal{C}}^{(t-1)} \|} \tag{4}$$

When $\Delta^{(t)} > \gamma$ (a shift threshold; $\gamma = 0.4$ in our experiments), all previously deactivated agents are reactivated for round $t+1$. The system uses each agent's *last valid output* as a proxy for novelty computation, avoiding new token generation for inactive agents. The overall framework is illustrated in Fig. 1.

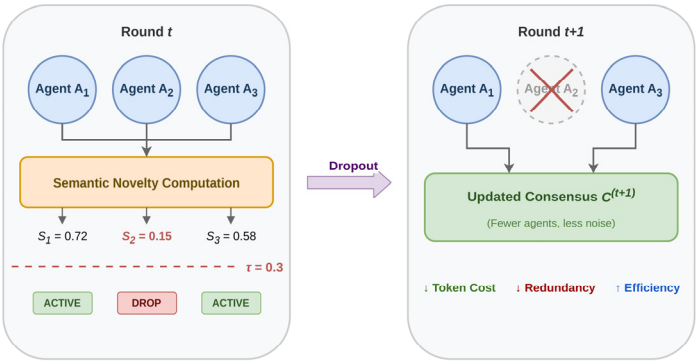


Fig. 1. Simplified overview of the AgentDropout framework. In Round t, the semantic novelty score $\mathcal{S}_i$ of each agent is computed relative to the current consensus. Agents with $\mathcal{S}_i < \tau$ (e.g., A_2 with $\mathcal{S}_2 = 0.15$) are deactivated. In Round $t+1$, only active agents (A_1, A_3) participate. The full logic includes adaptive threshold decay (Eq. 3) and consensus-shift-triggered reactivation (Eq. 4).

This reactivation policy is aligned with persistent-memory modeling and policy optimization for long-horizon LLM agents [43]. Retaining each agent's last valid output supplies a lightweight state representation that can be reconsidered when the consensus changes. The stored output allows the controller to reassess participation without paying the full cost of regenerating every inactive trajectory. In this way, temporary deactivation remains reversible rather than discarding potentially useful context.

E. Consensus Construction

After all active agents produce their outputs at round t, the consensus $\mathcal{C}^{(t)}$ is constructed by concatenating these outputs and prompting a designated summarizer (one of the active agents, selected round-robin) to produce a unified summary. This summary serves as both the representation against which novelty scores are computed and the shared context for the next round. We acknowledge that the summarizer's own perspective may influence the consensus (see Section VI).

Consensus construction also incorporates a conflict-aware view of evidence routing [44]. When the available signals disagree, the summary should preserve the distinction between supported agreement and unresolved conflict rather than forcing a premature unified answer. The shared representation therefore serves as both a compression of active outputs and a control signal for subsequent participation decisions. This separation allows confidence-sensitive routing to complement the novelty score without replacing the group deliberation protocol.

We further adopt a structure-grounded view of long-context reasoning [45]. The round-level summary is treated as an organized representation of the deliberation history, not merely a shorter transcript. Maintaining explicit relationships among active outputs, the current consensus, and the stored outputs of inactive agents supports faithful comparison across rounds. This structured context is especially important when an agent is reintroduced after the group state has changed.

IV. Experiments

A. Benchmarks and Setup

We evaluate on three benchmarks: **GSM8K** (1,319 grade-school math problems; consensus = same numerical answer), **StrategyQA** (2,290 yes/no commonsense questions; consensus = same yes/no answer), and **HumanEval** (164 Python programming problems; consensus = all programs passing the same example assertions). For HumanEval, the final answer is selected by the summarizer and evaluated via the official hidden test suite (pass@1).

We follow the multi-agent debate protocol: each agent independently generates an initial response, then iteratively refines it for up to $T = 6$ rounds or until consensus. We use GPT-3.5-Turbo (`gpt-3.5-turbo-0125`) as the backbone with temperature $= 0.7$ and top-$p = 0.95$. Sentence embeddings use `all-MiniLM-L6-v2`. Parameters: $\tau_{\text{base}} = 0.3$, $\alpha = 0.5$, $\gamma = 0.4$ (tuned on a held-out 200-problem GSM8K validation split), $N_{\min} = 2$. All experiments were

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conducted in January 2025 and repeated **3 times**; we report mean ± standard deviation.

B. Baselines

We compare against: **Single Agent** with chain-of-thought prompting, **Self-Consistency** (5 reasoning paths, majority vote), **Fixed N-Agent Debate** ($N \in \{3,5,7\}$), and **MetaGPT** for code generation. AgentDropout is initialized with $N = 5$ agents. For MetaGPT, we implement a simplified controlled variant using the same backbone model and unified token-counting protocol.

C. Evaluation Metrics

We report: (1) **Accuracy** (exact-match for GSM8K/StrategyQA, pass@1 for HumanEval), (2) **Total token consumption** (input + output tokens across all rounds and agents), (3) **Avg. active agents**, and (4) **Rounds to consensus**.

TABLE I. MAIN RESULTS ACROSS THREE BENCHMARKS (MEAN ± STD OVER 3 INDEPENDENT RUNS). TOKEN COUNTS ARE IN THOUSANDS (K). **AVG. ACT.** DENOTES AVERAGE ACTIVE AGENTS. AGENTDROPOUT IS INITIALIZED WITH 5 AGENTS. BEST DEBATE-BASED RESULTS PER BENCHMARK ARE **BOLDED**.

Method	Acc.(%)	Tok.(K)	Avg.Act.	Rnds
GSM8K				
Single Agent	$71.2_{\pm 0.8}$	$12.4_{\pm 0.3}$	1.0	–
Self-Consistency	$78.5_{\pm 0.6}$	$58.6_{\pm 1.4}$	–	–
Fixed 3-Agent	$80.5_{\pm 0.9}$	$36.8_{\pm 1.5}$	3.0	$3.6_{\pm 0.2}$
Fixed 5-Agent	$79.4_{\pm 1.1}$	$63.2_{\pm 2.4}$	5.0	$3.9_{\pm 0.2}$
Fixed 7-Agent	$77.8_{\pm 1.3}$	$89.7_{\pm 3.1}$	7.0	$4.2_{\pm 0.3}$
AgentDropout	**81.3**$_{\pm 0.7}$	**35.7**$_{\pm 1.2}$	$2.8_{\pm 0.2}$	**3.1**$_{\pm 0.2}$
StrategyQA				
Single Agent	$68.4_{\pm 0.7}$	$10.8_{\pm 0.2}$	1.0	–
Self-Consistency	$74.6_{\pm 0.5}$	$51.2_{\pm 1.2}$	–	–
Fixed 3-Agent	$77.0_{\pm 0.8}$	$32.1_{\pm 1.3}$	3.0	$3.4_{\pm 0.2}$
Fixed 5-Agent	$76.1_{\pm 1.0}$	$58.4_{\pm 2.1}$	5.0	$3.7_{\pm 0.2}$
Fixed 7-Agent	$73.8_{\pm 1.2}$	$82.1_{\pm 2.8}$	7.0	$4.0_{\pm 0.3}$
AgentDropout	**77.6**$_{\pm 0.6}$	**33.2**$_{\pm 1.1}$	$2.6_{\pm 0.2}$	**2.9**$_{\pm 0.2}$
HumanEval (pass@1)				
Single Agent	$65.8_{\pm 1.2}$	$15.2_{\pm 0.4}$	1.0	–
Fixed 3-Agent	$75.2_{\pm 1.4}$	$44.3_{\pm 1.8}$	3.0	$3.8_{\pm 0.3}$
Fixed 5-Agent	$74.4_{\pm 1.6}$	$74.5_{\pm 2.9}$	5.0	$4.1_{\pm 0.3}$
Fixed 7-Agent	$72.6_{\pm 1.5}$	$105.3_{\pm 3.8}$	7.0	$4.5_{\pm 0.3}$
MetaGPT	$74.4_{\pm 1.3}$	$68.2_{\pm 2.5}$	–	–
AgentDropout	**75.6**$_{\pm 1.1}$	**46.2**$_{\pm 1.6}$	$3.1_{\pm 0.2}$	**3.3**$_{\pm 0.2}$

V. RESULTS AND ANALYSIS

A. Main Results

Table I presents the main results. On GSM8K, AgentDropout achieves 81.3% accuracy, exceeding the fixed 5-agent baseline (79.4%) while consuming 43.5% fewer tokens. On StrategyQA, AgentDropout reaches 77.6%, matching or slightly exceeding fixed-agent baselines. On HumanEval, AgentDropout achieves 75.6% pass@1 with 38.0% fewer tokens than the fixed 5-agent baseline and 32.3% fewer than MetaGPT. The 7-agent fixed baseline tends to underperform the 3-agent configuration, corroborating that larger agent teams do not guarantee better outcomes. AgentDropout sidesteps this by starting with 5 agents and progressively pruning redundant contributors; the average active count stabilizes around 2.6–3.1.

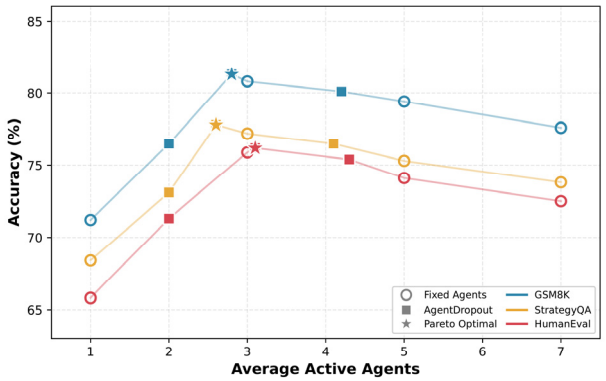


Fig. 2. Average active agent count vs. accuracy across three benchmarks. Hollow circles: fixed-agent baselines ($N = 1,3,5,7$). Filled squares: AgentDropout with different τ_{base} settings. Stars: Pareto-optimal points. Each point shows the mean over 3 independent runs; error bars are omitted for clarity (standard deviations are ≤1.6%).

B. Pareto Analysis of Active Agents vs. Accuracy

Fig. 2 plots accuracy against the average number of active agents. Fixed-agent baselines show a characteristic pattern: accuracy peaks around 3 agents and declines as redundancy accumulates. AgentDropout configurations consistently occupy the upper-left region (higher accuracy, fewer agents). The Pareto-optimal points correspond to $\tau_{\text{base}} = 0.3$ with an average of 2.6–3.1 active agents, suggesting that around 3 effective agents is a favorable operating region.

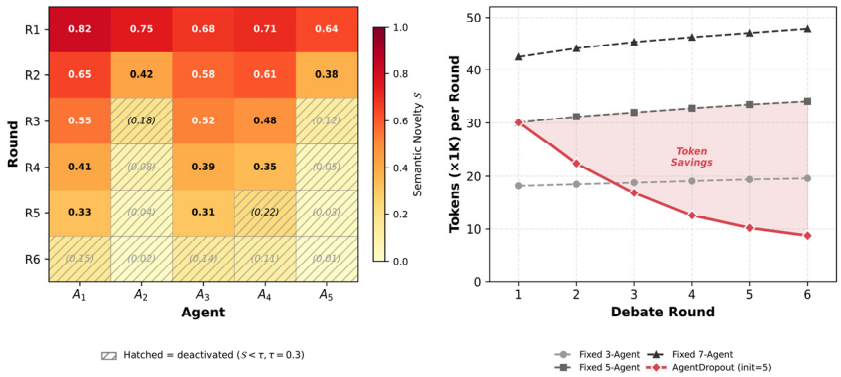


Fig. 3. **Left**: Heatmap of semantic novelty scores $\mathcal{S}_i^{(t)}$ for a single representative 5-agent GSM8K example. Hatched cells indicate deactivated agents ($\mathcal{S} < \tau$). **Right**: Average per-round token consumption across the full GSM8K test set (mean over 3 runs). AgentDropout (red) shows a sharply decreasing token profile, with the shaded area indicating cumulative savings relative to fixed 5-agent debate.

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TABLE II. ABLATION STUDY ON GSM8K (MEAN ± STD OVER 3 INDEPENDENT RUNS). EACH ROW REMOVES ONE COMPONENT FROM THE FULL AGENTDROPOUT SYSTEM.

Variant	Acc.(%)	Tokens(K)
Full AgentDropout	$81.3_{\pm 0.7}$	$35.7_{\pm 1.2}$
– Adaptive τ (fixed threshold)	$79.8_{\pm 0.9}$	$31.4_{\pm 1.0}$
– $N_{\min}$ constraint	$79.0_{\pm 1.1}$	$28.9_{\pm 1.4}$
– Novelty-based (random deact.)	$78.5_{\pm 1.3}$	$35.9_{\pm 1.1}$
– Reactivation mechanism	$80.3_{\pm 0.8}$	$33.6_{\pm 1.0}$

C. Novelty Score Dynamics and Token Savings

Fig. 3 (left) presents a heatmap of semantic novelty scores for a single representative 5-agent GSM8K example. In early rounds, most agents exhibit high novelty. As deliberation progresses, agents A_2 and A_5 rapidly converge toward consensus and are deactivated by round 3, while A_1, A_3, and A_4 maintain higher novelty longer. Fig. 3 (right) shows that AgentDropout exhibits a sharply decreasing per-round token profile compared to fixed baselines, accounting for the 38.0–43.5% total reduction.

D. Ablation Study

Table II presents ablation results on GSM8K. Removing adaptive threshold decay reduces accuracy by 1.5 pp, as useful agents are prematurely eliminated when diversity matters most. Disabling $N_{\min}$ leads to a 2.3 pp drop (occasional degeneration to a single agent). Random deactivation degrades accuracy by 2.8 pp, confirming the importance of principled agent selection. Removing reactivation causes a modest 1.0 pp drop, consistent with consensus shifts being relatively infrequent ($\sim 8\%$ of problems).

VI. DISCUSSION AND LIMITATIONS

Our results suggest that not all agents need to participate throughout the entire deliberation. Early rounds benefit from full participation to maximize reasoning diversity, while later rounds can tolerate reduced agent counts as the group converges.

Several limitations should be noted. First, our semantic novelty metric relies on sentence-level embeddings, which may not capture fine-grained logical differences—two solutions with identical answers but different reasoning chains could receive insufficiently differentiated scores. Second, for HumanEval, example-assertion-based consensus is only a weak proxy for functional equivalence. Third, the threshold parameters were tuned on a single GSM8K validation set and may require task-specific adjustment. Fourth, the round-robin summarizer's perspective may bias the consensus against which other agents are scored. Fifth, our experiments use homogeneous agents (all GPT-3.5-Turbo); heterogeneous model ensembles may exhibit different dynamics. Finally, accuracy differences between AgentDropout and the strongest fixed baselines are modest (0.4–1.9 pp); larger-scale repetition would further strengthen the conclusions.

VII. CONCLUSION

We have presented AgentDropout, a dynamic mechanism for reducing redundancy in multi-agent LLM collaboration by selectively deactivating low-contribution agents based on semantic novelty scoring. Experiments on GSM8K, StrategyQA, and HumanEval demonstrate that AgentDropout achieves accuracy on par with fixed 5-agent debate while reducing token consumption by 38.0–43.5%. The Pareto analysis suggests that around 3 effective agents is a favorable operating region. Future work should explore more expressive contribution measures, heterogeneous model teams, and more diverse task categories.

REFERENCES

[1] T. Guo et al., "Large language model based multi-agents: A survey of progress and challenges," in Proceedings of the 33rd international joint conference on artificial intelligence (IJCAI), 2024, pp. 8048–8057.

[2] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch, "Improving factuality and reasoning in language models through multiagent debate," in Proceedings of the 41st international conference on machine learning (ICML), PMLR, 2024.

[3] T. Liang et al., "Encouraging divergent thinking in large language models through multi-agent debate," in Proceedings of the 2024 conference on empirical methods in natural language processing (EMNLP), 2024, pp. 17889–17904.

[4] S. Hong et al., "MetaGPT: Meta programming for a multi-agent collaborative framework," in The twelfth international conference on learning representations (ICLR), 2024.

[5] C. Qian et al., "ChatDev: Communicative agents for software development," in Proceedings of the 62nd annual meeting of the association for computational linguistics (ACL), 2024.

[6] Q. Wu et al., "AutoGen: Enabling next-gen LLM applications via multi-agent conversation," arXiv preprint arXiv:2308.08155, 2023.

[7] A. P. Smit, N. Grinsztajn, P. Duckworth, T. D. Barrett, and A. Pretorius, "Should we be going MAD? A look at multi-agent debate strategies for LLMs," in Proceedings of the 41st international conference on machine learning (ICML), PMLR, 2024, pp. 45883–45905.

[8] X. Wang et al., "Self-consistency improves chain of thought reasoning in language models," in The eleventh international conference on learning representations (ICLR), 2023.

[9] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," Journal of Machine Learning Research, vol. 15, no. 56, pp. 1929–1958, 2014.

[10] K. Cobbe et al., "Training verifiers to solve math word problems," arXiv preprint arXiv:2110.14168, 2021.

[11] M. Geva, D. Khashabi, E. Segal, T. Khot, D. Roth, and J. Berant, "Did aristotle use a laptop? A question answering benchmark with implicit reasoning strategies," Transactions of the Association for Computational Linguistics, vol. 9, pp. 346–361, 2021.

[12] M. Chen et al., "Evaluating large language models trained on code," arXiv preprint arXiv:2107.03374, 2021.

[13] J. C.-Y. Chen, S. Saha, and M. Bansal, "ReConcile: Round-table conference improves reasoning via consensus among diverse LLMs," in Proceedings of the 62nd annual meeting of the association for computational linguistics (ACL), 2024, pp. 7066–7085.

[14] J. Y. Su and X. Yan, "AI-Assisted Emotional Expression Analysis of Children's Drawings for School-Based Art Therapy Support," 2026, doi: 10.13140/RG.2.2.20510.09289.

[15] Z. Peng, S. Liu, X. Meng, Z. Su, and N. Sang, "A Schema-Aware Large Language Model for XBRL Tagging and Cross-Statement Consistency Checking," 2026, doi: 10.13140/RG.2.2.19518.86080.

[16] X. Liu, C. Y. Hsieh, Z. Wang, Y. Liu, and Z. Xiao, "Multi-Agent Collaborative Reasoning and Role Division Optimization for Emotion Recognition under Complex Context Understanding," 2026, doi: 10.13140/RG.2.2.27883.71208.

[17] Y. Zhao, Y. Li, Y. Wang, Y. Nie, Y. Lu, and N. Chen, "Conservative Risk-Sensitive Reinforcement Learning for Reliable Decision-Making Under Uncertainty," in 2026 9th International Symposium on Big Data and Applied Statistics (ISBDAS), Mar. 2026, pp. 102-107.

[18] J. Liao, F. Chang, T. Xia, Z. Huang, X. Huang, and Y. Zhou, "Lightweight Edge-Cloud Cooperative Inference for Small and Medium Models under WAN Constraints," in 2026 5th International Conference on Electronic

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Information Engineering, Big Data and Computer Technology (EIBDCT), Apr. 2026, pp. 419-424.

[19] C. Zhang, H. Zhu, A. Zhu, J. Liao, Y. Xiao, and Z. Zhang, "Deep Learning Approach for Protocol Anomaly Detection Using Status Code Sequences," in Proceedings of the 2026 International Conference on Generative Artificial Intelligence and Education, Feb. 2026, pp. 10-15.

[20] R. Meng, S. Huang, X. Zhang, Y. Yang, K. Zeng, and Z. Huang, "Constraint-Consistent Skill Composition for Reliable Zero-Shot Task Generalization in LLM Agents," in 2026 9th International Symposium on Big Data and Applied Statistics (ISBDAS), Mar. 2026, pp. 620-626.

[21] J. Huang, J. Zhan, Q. Wang, J. Jia, and B. Zhang, "Stable Fault Diagnosis under Data Imbalance via Self-Supervised Learning in Industrial IoT," in 2026 5th International Conference on Electronics Technology and Artificial Intelligence (ETAI), Mar. 2026, pp. 424-429.

[22] C. Wen, A. Zhu, R. Long, H. Huang, J. Jiang, and C. S. Lee, "CalibJudge: Calibrated LLM-as-a-Judge for Multilingual RAG with Uncertainty-Aware Scoring," in Proceedings of the 2026 International Conference on Artificial Intelligence and Control, Feb. 2026, pp. 661-667.

[23] Y. Nie, J. Wang, R. Yan, Y. Wang, Z. Ma, and Y. Wu, "Graph-Based Financial Fraud Detection with Calibrated Risk Scoring and Structural Regularization," in Proceedings of the 2026 International Conference on AI Decision-Making and Management, Mar. 2026, pp. 169-176.

[24] H. Zhuang, W. Huang, N. Lyu, J. Kou, W. Huang, and R. Wei, "TokenPool-Scheduler: Token-Level GPU Pooling and Resource Slicing for Multi-Model Co-Location," in 2026 IEEE 8th International Conference on Communications, Information System and Computer Engineering (CISCE), Mar. 2026, pp. 1630-1635.

[25] W. Wang, Y. Zhang, Z. Pan, Z. Ke, Y. Sun, and Y. Yang, "Drift-Aware Agentic ETL for Schema-Evolving Data Lakes," in 2026 International Conference on Generative Artificial Intelligence and Information Security (GAIIS), Mar. 2026, pp. 728-733.

[26] Z. Ma, Z. Ke, L. Tang, Z. Bian, and X. Ren, "DQ-MCTR: Data Quality-Aware Multimodal Click-Through Rate Prediction for Online Advertising," 2026, doi: 10.13140/RG.2.2.15862.77127.

[27] L. Tang, Y. Mei, X. Ren, Y. Ke, Z. Ke, and Z. Bian, "DriftMender: LLM-Guided Semantic Repair for Schema Drift in Real-Time ETL Pipelines," 2026, doi: 10.13140/RG.2.2.19248.52483.

[28] Y. Mei, Y. Ke, S. Li, and Q. Cheng, "Attention-Based Representation Learning for XBRL-Guided Corporate Revenue Forecasting," 2026, doi: 10.13140/RG.2.2.19002.76489.

[29] L. Ma, T. Song, R. Long, L. Mao, and Z. Gu, "Evidence-Gated Memory Writing for Personalized Large Language Model Agents," 2026, doi: 10.13140/RG.2.2.10244.69767.

[30] S. Pi, S. Zhang, and S. Shawulieti, "CacheRoute: KV-Cache-Aware Routing for LLM Backend Services on Kubernetes," 2026, doi: 10.13140/RG.2.2.34191.55206.

[31] S. Lin, Y. Lu, Z. Peng, S. Sun, Y. Wang, and M. Wang, "XBRL-Aware LLM Agents for Reliable Financial Statement Analysis: Structured Reasoning and Anti-Hallucination Verification," 2026, doi: 10.13140/RG.2.2.17602.75206.

[32] X. Li, X. Liu, Z. Wang, Z. Xiao, C. Y. Hsieh, and Y. Liu, "DeepServeCB: SLO-Aware and Cost-Aware Elastic Scheduling for Serverless Multi-Tenant LLM Inference," in 2026 International Conference on Generative Artificial Intelligence and Information Security (GAIIS), Mar. 2026, pp. 90-95.

[33] Y. Wang, Y. Sun, Y. Yang, Y. Zhang, W. Wang, and Z. Pan, "A Data-Centric Sequential Modeling Framework for Session-Level Decision Intelligence," in 2026 IEEE 8th International Conference on Communications, Information System and Computer Engineering (CISCE), Mar. 2026, pp. 1613-1618.

[34] C. Su, L. Jiang, and S. Pi, "Sentiment Classification of Chinese Railway Review Text Based on Multi-Feature Fusion Gated Recurrent Unit," in 2021 International Conference on Information Science, Parallel and Distributed Systems (ISPDS), Aug. 2021, pp. 197-202.

[35] S. Li, Y. Wang, Y. Xing, and M. Wang, "Mitigating Correlation Bias in Advertising Recommendation via Causal Modeling and Consistency-Aware Learning," in Proceedings of the 2025 6th International Conference on Computer Science and Management Technology, Dec. 2025, pp. 585-589.

[36] Z. Ke, "Right-Sizing Cloud Data Warehouses Under Asymmetric Loss via One-Sided Conformal Prediction," IEEE Access, vol. 14, pp. 94748-94763, 2026.

[37] Y. Xing, Y. Deng, H. Liu, M. Wang, Y. Zi, and X. Sun, "Contrastive Learning-Based Dependency Modeling for Anomaly Detection in Cloud Services," in 2025 6th International Conference on Information Science, Parallel and Distributed Systems (ISPDS), London, U.K., 2025, pp. 219-224, doi: 10.1109/ISPDS67367.2025.11390903.

[38] Y. Xing, M. Wang, Y. Deng, H. Liu, and Y. Zi, "Explainable Representation Learning in Large Language Models for Fine-Grained Sentiment and Opinion Classification," in 2025 7th International Academic Exchange Conference on Science and Technology Innovation (IAECST), Guangzhou, China, 2025, pp. 11-16, doi: 10.1109/IAECST68792.2025.11415236.

[39] Y. Li et al., "Improving multi-agent debate with sparse communication topology," in Findings of the association for computational linguistics: EMNLP 2024, 2024, pp. 7281–7294.

[40] J. Wei et al., "Chain-of-thought prompting elicits reasoning in large language models," in Advances in neural information processing systems (NeurIPS), 2022, pp. 24824–24837.

[41] W. Ma, K. Yang, Z. Liu, R. Jiang, and Y. Jiang, "VerifyRCA: LLM-Guided but Evidence-Verified Root Cause Analysis for Microservice Backends," 2026, doi: 10.13140/RG.2.2.28829.88803.

[42] F. Chen, B. Chen, Y. Tang, Z. Hu, J. Zhang, and Y. Zhan, "ErrorChain: Error Cascade Detection and Early Truncation in Multi-Step Agents," 2026, doi: 10.13140/RG.2.2.21919.11688.

[43] Y. Ying, L. Wang, P. Hu, and T. Ding, "Persistent Memory Modeling and Policy Optimization for Large Language Model Agents in Long-Horizon Tasks," 2026, doi: 10.13140/RG.2.2.23282.26560.

[44] R. Fang, J. Zhou, Z. Zhu, R. Hu, and Y. Zheng, "Confidence-Conflict Routing Between Parametric Memory, RAG, and Long-Context LLMs," 2026, doi: 10.13140/RG.2.2.23862.25927.

[45] J. Zhou, Z. Zhu, A. Zhu, S. Wang, and Y. Zheng, "Structure-Grounded Document Agents for Faithful Long-Context Reasoning," 2026, doi: 10.13140/RG.2.2.25574.38723.

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